Phase-Aware Dense Rewards for Compute-Constrained Vision-Language-Action Reinforcement Learning

Kamil Nuriev and Egor Novikov
Innopolis University, Innopolis, Russia
{k.nuriev, eg.novikov}@innopolis.university

***Abstract*—Online reinforcement learning (RL) for Vision-Language-Action (VLA) policies is expensive because each update requires simulator interaction and inference through a large multimodal model. Binary outcome rewards further weaken credit assignment when a rollout batch contains few successful trajectories. We extend SimpleVLA-RL with simulator-derived, phase-aware dense shaping for OpenVLA-OFT. Supported pick-and-place tasks are decomposed into reach, grasp, lift, move, and place phases. The reward credits newly achieved within-phase progress and phase transitions while retaining the original verifier success reward as the task objective. We evaluate the method on LIBERO-Spatial using two NVIDIA L40 GPUs and an aggregate rollout batch of 32. From a common 61.0% initial policy, the primary dense configuration reaches 65.3% success, compared with 61.7% for terminal-only training. In a controlled low-success task experiment, dense and terminal training finish at 18% and 11%, respectively, from the same 10% initialization. The dense run also assembles accepted training batches faster and completes more optimizer updates over the observed training interval. The results show that phase-aware dense feedback produces a larger observed improvement than terminal-only reward under compute-constrained VLA RL.**

***Index Terms*—vision-language-action models, reinforcement learning, reward shaping, GRPO, robotic manipulation, LIBERO**

I. Introduction

Vision-Language-Action (VLA) models map visual observations and language instructions to robot actions. OpenVLA established an open-source VLA foundation model for robot control [1], and OpenVLA-OFT introduced an efficient fine-tuning recipe with parallel action decoding and action chunking [2]. SimpleVLA-RL extends these policies with online reinforcement learning based on Group Relative Policy Optimization (GRPO) and binary task verification [3], [4].

Online VLA RL remains computationally demanding: every policy update requires parallel environment rollouts, repeated multimodal inference, reward computation, and optimization of a large policy. The published SimpleVLA-RL setup uses eight NVIDIA A800 80GB GPUs [3]. We study whether informative intermediate rewards can improve training with two NVIDIA L40 GPUs and an aggregate rollout batch of 32.

Binary success rewards are aligned with task completion, but all unsuccessful trajectories receive the same outcome even when they make different amounts of progress. This is restrictive for low-success tasks, where a small batch may contain little or no outcome contrast. Dense shaping can provide intermediate credit, although arbitrary shaping can alter the optimization objective or reward repeated, nonproductive behavior [5].

We add phase-aware dense shaping to SimpleVLA-RL for LIBERO-Spatial [6]. The implementation extracts simulator state, divides supported pick-and-place behavior into five phases, rewards only progress beyond the best state already reached in the active phase, and clips shaping independently of terminal reward. The contributions are:

- a phase-aware reward combining monotonic within-phase progress and phase transitions with the original terminal verifier reward;
- a controlled comparison showing a larger observed LIBERO-Spatial improvement with dense shaping than with terminal-only reward under a two-L40, batch-32 setup;
- a low-success single-task analysis of validation success and accepted-batch collection during training.

II. Background

A. VLA Reinforcement Learning

OpenVLA transfers pretrained vision-language representations to robot control [1]. OpenVLA-OFT improves supervised fine-tuning through continuous action representation, parallel action decoding, action chunking, and an L_1 action objective [2]. The present implementation starts from an OpenVLA-OFT checkpoint but optimizes the likelihood of discrete action tokens with PPO/GRPO. The decoded token sequence is then unnormalized into a continuous robot action chunk.

SimpleVLA-RL adapts the veRL distributed RL stack to online VLA rollouts and environment verification [3]. GRPO estimates relative advantages within groups of responses sampled for the same prompt and avoids a separate value model [4]. In robotics, each response is a trajectory and the verifier reports whether the environment goal was achieved. A group of uniformly unsuccessful trajectories provides no binary-reward contrast.

Reward shaping is a standard response to sparse credit assignment. Potential-based shaping can preserve the optimal policy under specific assumptions [5]; our phase-aware reward is not claimed to satisfy that theorem. It is an engineering

mechanism that limits repeated reward through monotonic progress tracking, bounded shaping, and a terminal-success anchor. Policy updates use LoRA [7].

III. Method

A. Phase-Aware Reward

Supported LIBERO pick-and-place tasks are represented by five ordered phases: reach, grasp, lift, move, and place. The implementation derives object, target, gripper, and action state from the simulator. Each parallel environment maintains an independent phase tracker.

Let $p_t \in [0, 1]$ be progress in the active phase and B_t the best progress previously recorded for that phase. The rewarded progress is

$$\Delta p_t^+ = \max(p_t - B_t, 0). \tag{1}$$

The tracker resets its progress baseline when it advances to a new phase. Returning to an already reached state therefore receives no additional progress reward.

Let c_t be the number of newly completed phases and $y_t \in \{0, 1\}$ the original verifier terminal-success indicator. The shaping component is

$$r_t^{\text{shape}} = \text{clip}\big(w_p \Delta p_t^+ + w_c c_t, -C, C\big). \tag{2}$$

Additive mode combines this shaping signal with the original verifier reward,

$$r_t = \lambda y_t + r_t^{\text{shape}}. \tag{3}$$

The parameter `clip_dense_reward` supplies C. It clips only progress and phase-transition shaping; it does not clip the original terminal verifier reward. No separate terminal term is added by the dense reward in the reported experiments.

B. Mixed-Outcome Filtering

Reward clipping and rollout filtering are distinct. For a prompt with n sampled trajectories and terminal outcomes Y_i, a group is retained when

$$L \le \frac{1}{n} \sum_{i=1}^{n} Y_i \le U. \tag{4}$$

Bounds that exclude zero select groups containing a minimum fraction of successful trajectories. Groups outside the configured interval are not passed to the policy update.

IV. Experimental Setup

A. Model and Compute

Experiments use an OpenVLA-OFT checkpoint with LoRA rank and alpha of 16 applied to `llm-projector`. The benchmark is LIBERO-Spatial, which contains ten spatial-relation manipulation tasks [6]. Training uses two NVIDIA L40 GPUs. The reported configurations use an aggregate rollout batch of 32 across the two rollout workers. Gradient checkpointing and FSDP parameter, gradient, and optimizer offloading reduce memory demand. All comparisons start from the same policy, with an aggregate validation success of 0.610.

TABLE I
LIBERO-Spatial reward ablation. Δ is relative to the common 0.610 initialization.

Configuration	Success	Δ (pp)
Primary dense	**0.653**	**+4.3**
Filtered dense $[.1, .9]$	0.647	+3.7
Phase shaping, $\lambda = 1$	0.623	+1.3
Terminal only, $\lambda = 5$	0.617	+0.7
Dense shaping, $\lambda = 1$	0.610	0.0

B. Reward Configurations

The terminal control disables dense reward and uses verifier coefficient $\lambda = 5$. The primary dense configuration uses $w_p = 0.2$, $w_c = 0.3$, shaping clip $C = 0.05$, and $\lambda = 5$. A mixed-outcome filtered run uses the same reward and group-accuracy bounds $[0.1, 0.9]$. Two weaker-terminal ablations use $\lambda = 1$: dense shaping with $w_p = w_c = 0.05$, and phase-only shaping with $w_c = 0.2$. Both use $C = 0.05$. All configurations retain the original verifier reward as their only terminal term.

C. Evaluation

Every aggregate result uses evaluation episodes, balanced across the ten LIBERO-Spatial tasks. Success is the mean binary environment outcome. The same initial policy and evaluation protocol are used across reward ablations. We report the final recorded evaluation for each configuration rather than selecting a checkpoint independently for each run.

V. Results

A. LIBERO-Spatial Reward Ablation

Table I and Fig. 1 summarize the aggregate comparison. Primary dense shaping improves success from 0.610 to 0.653, a gain of 4.3 percentage points. Terminal-only training reaches 0.617, a gain of 0.7 points. The final dense result is therefore 3.6 points higher than the terminal control. Mixed-outcome filtered dense reaches 0.647, while phase-only shaping and dense shaping with a weaker terminal coefficient reach 0.623 and 0.610, respectively.

The two strongest results use dense phase-aware feedback, supporting improved observed training outcome under the tested compute budget. The terminal scale also appears important: primary dense shaping with verifier coefficient $\lambda = 5$ improves by 4.3 points, whereas the weaker-terminal dense configuration with $\lambda = 1$ does not improve on initialization. Because these configurations also use different shaping weights, this comparison does not isolate the causal effect of λ; it instead shows that dense shaping does not remove the need for a sufficiently strong task-completion objective.

B. Low-Success Task Study

We additionally restrict both training and validation to LIBERO-Spatial `task_1`, whose initial success is 0.10. Both conditions use a rollout-group accuracy filter of $[0.1, 1.0]$; the only intended experimental difference is the reward. Table II reports the initial, intermediate, and final evaluations.

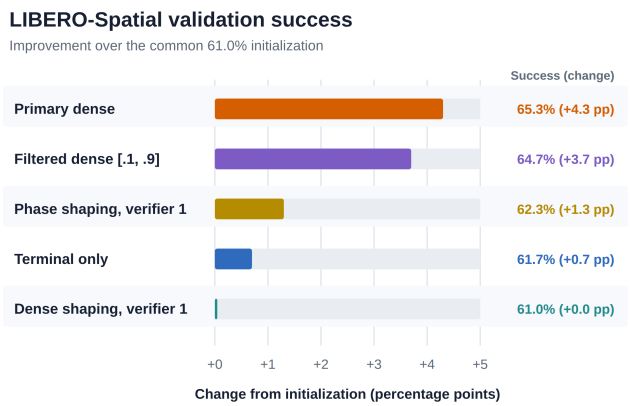


Fig. 1. LIBERO-Spatial validation success across reward ablations. Bars show improvement from the common 61.0% initialization; labels report final success and absolute change.

TABLE II
SINGLE-TASK SUCCESS WITH GROUP-ACCURACY FILTERING $[0.1, 1.0]$.

Evaluation	Terminal only	Dense
Initial	0.10	0.10
Intermediate	0.08	0.12
Final	0.11	**0.18**
Change from initial	+0.01	**+0.08**

Dense reward improves by 8 percentage points from initialization, compared with 1 point for terminal reward, and finishes 7 points higher.

The training logs show that the dense condition assembles accepted batches faster than the terminal-only condition. Consequently, the dense run completes more optimizer updates over the observed training interval. This throughput difference accompanies the larger validation improvement and indicates that intermediate reward helped the policy reach acceptable rollout groups more consistently on the low-success task.

VI. Discussion

The principal result is a larger observed validation improvement under constrained compute. Starting from 61.0%, the primary dense configuration gains 4.3 points, while terminal-only training gains 0.7 points. The reward structure provides a plausible credit-assignment explanation: terminal reward identifies completed tasks, whereas monotonic progress and phase transitions distinguish partially successful trajectories that would otherwise share the same zero outcome reward.

The ablations also show that dense reward is not automatically beneficial. Its effect depends on progress definitions, component weights, terminal scale, and the rollout selection policy. The weaker-terminal result suggests that intermediate feedback should complement rather than replace a strong terminal objective. Mixed-outcome filtering remains a separate mechanism: it controls which groups reach the optimizer, while dense shaping changes the reward assigned to collected trajectories.

VII. Conclusion

We introduced phase-aware dense reward shaping for compute-constrained SimpleVLA-RL. On LIBERO-Spatial with two NVIDIA L40 GPUs and an aggregate rollout batch of 32, the primary dense configuration improves from 61.0% to 65.3%, compared with 61.7% for terminal-only training. A controlled low-success task study shows the same ordering, with final success of 18% for dense and 11% for terminal reward. In that experiment, the dense run also assembles accepted batches faster and completes more optimizer updates during the observed training interval. The ablations further indicate that dense feedback works best as a complement to a sufficiently strong terminal verifier reward. These results support phase-aware dense feedback as a practical way to improve small-batch VLA RL.

References

[1] M. J. Kim *et al.*, "OpenVLA: An open-source vision-language-action model," in *Proc. Conf. Robot Learning (CoRL)*, vol. 270, 2024, pp. 2679–2713.

[2] M. J. Kim, C. Finn, and P. Liang, "Fine-tuning vision-language-action models: Optimizing speed and success," *arXiv preprint arXiv:2502.19645*, 2025.

[3] H. Li *et al.*, "SimpleVLA-RL: Scaling VLA training via reinforcement learning," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2026.

[4] Z. Shao *et al.*, "DeepSeekMath: Pushing the limits of mathematical reasoning in open language models," *arXiv preprint arXiv:2402.03300*, 2024.

[5] A. Y. Ng, D. Harada, and S. Russell, "Policy invariance under reward transformations: Theory and application to reward shaping," in *Proc. 16th Int. Conf. Machine Learning*, 1999, pp. 278–287.

[6] B. Liu *et al.*, "LIBERO: Benchmarking knowledge transfer for lifelong robot learning," in *Advances in Neural Information Processing Systems*, vol. 36, Datasets and Benchmarks Track, 2023.

[7] E. J. Hu *et al.*, "LoRA: Low-rank adaptation of large language models," in *Proc. Int. Conf. Learning Representations*, 2022.